

AMC 8 Crash Course • Block 1 • Numbers and Arithmetic •

Lessons 1.1–1.5

Theory, worked examples, and independent sets. Solutions to the independent sets are at the end of the file.

Lesson 1.1. Fractions: Comparing and Operations

THEORY

Comparing fractions. Three ways, fastest first: compare with one half (in $\frac{5}{9}$ the numerator is more than half the denominator, so it is greater than $\frac{1}{2}$); cross-multiplication ($\frac{2}{5}$ vs $\frac{3}{8}$: multiply each numerator by the other fraction's denominator — $2 \cdot 8 = 16$ against $3 \cdot 5 = 15$; the larger product belongs to the larger fraction, so $\frac{2}{5}$ wins); common denominators.

A fraction of a number means multiplication: $\frac{3}{5}$ of 45 is $45 \cdot \frac{3}{5} = 27$. Divide by the denominator first, then multiply. The reverse problem — a number from its fraction: if $\frac{2}{3}$ of a number is 18, then one third is 9, so the number is 27.

Adding and subtracting. Use the least common denominator: $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. Always reduce the answer.

“Part of what’s left” problems. Work step by step and watch WHAT each fraction is taken of: the second fraction is almost always taken of the remainder, not of the original number. That is the number one trap in fraction problems.

WORKED EXAMPLES

EXAMPLE 1 • COMPARING

Which is larger: $\frac{3}{7}$ or $\frac{4}{9}$?

→ Cross-multiply: $3 \cdot 9 = 27$ and $4 \cdot 7 = 28$. The larger product marks the larger fraction: $\frac{4}{9}$. The method always works, no common denominator needed.

EXAMPLE 2 • ADDITION

Compute $\frac{2}{3} + \frac{1}{4}$.

→ The least common denominator is 12: $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$. Adding numerators and denominators separately ($2+1$ over $3+4$) is the classic mistake.

EXAMPLE 3 • FRACTION OF A NUMBER

Find $\frac{3}{8}$ of 56.

→ $56/8 = 7$, then $7 \cdot 3 = \mathbf{21}$. Divide first, multiply after — the numbers stay small.

EXAMPLE 4 • PART OF THE REST

A box held 30 candies. First $\frac{2}{5}$ of all the candies were eaten, then $\frac{1}{3}$ of the rest. How many candies are left?

→ Step 1: $30 \cdot \frac{2}{5} = 12$ eaten, 18 left. Step 2: $18 \cdot \frac{1}{3} = 6$ eaten, **12** left. The trap is taking $1/3$ of 30: the second fraction applies to the remainder.

INDEPENDENT SET

1. Write $\frac{18}{24}$ in lowest terms.

2. Which is larger: $\frac{5}{8}$ or $\frac{7}{12}$?
3. Compute $\frac{5}{6} - \frac{1}{2}$.
4. Find $\frac{2}{9}$ of 63.
5. $\frac{2}{7}$ of a number equals 10. Find the number.
6. Compute $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$.
7. A shelf holds 24 books, and $\frac{5}{8}$ of them are fairy tales. How many books are not fairy tales?
8. Find a fraction with denominator 12 that is greater than $\frac{1}{3}$ but less than $\frac{1}{2}$.
9. ★ In a class, $\frac{3}{5}$ of the students are girls. After 4 new boys join, girls make up exactly half of the class. How many students were in the class at first?

Lesson 1.2. Percents

THEORY

A percent is one hundredth. $25\% = 25/100 = 1/4$. Worth memorizing: $50\% = 1/2$, $25\% = 1/4$, $20\% = 1/5$, $10\% = 1/10$, $75\% = 3/4$.

Fast mental math: find 10% (divide by 10) and build from it. 15% of 180: $10\% = 18$, $5\% = 9$, together 27.

A number from its percent. If 20% of a number is 12, then 10% is 6, so the number is 60. Always drop down to 10% or 1%. Two related questions: “what percent of 48 is 12?” — that is the fraction $12/48 = 1/4$ written as a percent (25%); “by what percent did the price change?” — always divide the change by the *old* value.

Discounts and markups. A 15% discount means 85% of the price remains. Two ways to compute: subtract the discount ($80 - 12$) or take the remainder directly ($80 \cdot 0.85$) — same answer.

WORKED EXAMPLES

EXAMPLE 1 • FAST MENTAL MATH

Find 15% of 240.

→ $10\% = 24$, half of that (5%) = 12. Together: $24 + 12 = 36$. No long multiplication: percents snap together from 10% like blocks.

EXAMPLE 2 • NUMBER FROM A PERCENT

30% of a number is 45. Find the number.

→ If 30% is 45, then 10% is 15. The whole number (100%) is $15 \cdot 10 = 150$.

EXAMPLE 3 • DISCOUNT

A sweater cost \$60 and was marked down 25%. What is the new price?

→ 75% of the price remains: $60 \cdot 3/4 = 45$. Or: the discount is $60/4 = 15$, so $60 - 15 = 45$.

EXAMPLE 4 • PERCENT AND THE REST

In a class, 40% of the students are girls, and there are 18 boys. How many students are in the class?

→ The boys are 60%. If $60\% = 18$, then $10\% = 3$, so the class has **30** students. The key move: switch from girls to boys, because the given number is about boys.

INDEPENDENT SET

1. Find 20% of 45.
2. What percent of 48 is 12?
3. Increase 80 by 15%.
4. 5% of a number is 7. Find the number.
5. A price dropped from \$250 to \$200. By what percent did it fall?
6. Tommy had \$200 and spent 35%. How much is left?
7. A school has 300 students, and 12% attend chess club. How many students is that?
8. After a 20% discount an item costs \$96. What was the price before the discount?
9. ★ A price was first raised by 25%, then lowered by $p\%$ — and it returned exactly to the original. What is p ?

Lesson 1.3. Ratios and Proportions

THEORY

A ratio means parts. If two numbers are in the ratio 3 : 5, picture 3 parts and 5 parts of the same size. The total is 8 parts: knowing the total, divide to get one part.

Splitting in a ratio: divide the total by the number of parts. 56 in the ratio 3 : 4 — one part is $56/7 = 8$, the numbers are 24 and 32.

A proportion is an equality of ratios. Recipes, maps, prices: if 3 cups of rice cook in 4 cups of water, then 9 cups of rice need three times the water. Find how many times one quantity grew and multiply the other by the same factor. And when ratios are chained through a shared quantity ($a:b$ and $b:c$), scale b to the same number in both — and the chain links up.

Unit rate. The easiest way to compare and rescale is the price of one item: 6 stickers cost 42 cents, so one costs 7, and ten cost 70.

WORKED EXAMPLES

EXAMPLE 1 • SPLITTING IN A RATIO

Split 72 in the ratio 3 : 5.

→ 8 parts in all, one part = $72/8 = 9$. The numbers: 27 and 45. Check: $27 + 45 = 72$. **27 and 45.**

EXAMPLE 2 • RECIPE

A recipe uses 3 tablespoons of sugar for every 2 cups of flour. How many tablespoons of sugar are needed for 8 cups of flour?

→ The flour grew fourfold ($8/2 = 4$), so the sugar does too: $3 \cdot 4 = 12$.

EXAMPLE 3 • MAP

A map has scale 1 : 50,000. What real distance corresponds to 4 cm on the map?

→ 4 cm on the map is $4 \cdot 50,000 = 200,000$ cm = **2 km**. The real work is the unit conversion; do it last.

EXAMPLE 4 • UNIT RATE

Six pencils cost 90 cents. How much do 10 such pencils cost?

→ One pencil: $90/6 = 15$ cents. Ten: **150 cents** (\$1.50).

INDEPENDENT SET

1. Split 45 in the ratio 2 : 7. What is the larger part?
2. Simplify the ratio 12 : 18.
3. Five identical apples weigh 400 grams. How much do 8 such apples weigh?
4. $a : b = 3 : 4$ and $b : c = 2 : 5$. Find $a : c$.
5. An audiobook narrator reads 180 words in 3 minutes. What is the rate in words per minute?
6. Find x if $3 : 4 = x : 20$.
7. A trail mix has raisins and peanuts in the ratio 2:3. There are 14 ounces of raisins. How many ounces does the whole mix weigh?
8. The ratio of boys to girls in a class is 4 : 5, and there are 3 more girls than boys. How many students are there in all?
9. ★ Two numbers are in the ratio 5 : 3. If 10 is subtracted from the larger, the ratio becomes 5 : 4. Find the sum of the original numbers.

Lesson 1.4. Averages

THEORY

The mean is the sum divided by the count. But problems are easier the other way round: **sum = mean × count**. Nearly every average problem is really a sum problem.

Find the sum before and after the change — and the problem solves itself. A number added? New sum minus old. Removed? Old minus new.

The missing number. Need an average of 90 across three quizzes, and the first two gave 88 and 86? The required sum is 270, you have 174 — the third must be 96. A useful shift: if every number grows by 3, the mean grows by exactly 3.

The median is the middle value of the sorted list. With an odd count it is the middle element; with an even count, the average of the two middle ones: for 3, 5, 9, 11 the median is $(5 + 9)/2 = 7$. Sorting is mandatory. On the AMC 8 the median comes up as often as the mean.

WORKED EXAMPLES

EXAMPLE 1 • VIA THE SUM

The mean of five numbers is 14. The number 26 is removed. What is the mean of the remaining four?

→ The sum was $5 \cdot 14 = 70$. It became $70 - 26 = 44$. Mean: $44/4 = 11$.

EXAMPLE 2 • DIRECT COMPUTATION

Test scores: 78, 85, 92. What is the mean?

→ $(78 + 85 + 92)/3 = 255/3 = 85$. Faster: notice 78 and 92 are symmetric around 85.

EXAMPLE 3 • THE MISSING TEST

Three tests scored 74, 82, and 79. What must the fourth score be for the four-test average to equal 80?

→ The required sum is $4 \cdot 80 = 320$. So far 235. The fourth test: $320 - 235 = 85$.

EXAMPLE 4 • MEDIAN

Find the median of 12, 3, 14, 7, 8.

→ Sort: 3, 7, 8, 12, 14. The middle element is **8**. Sorting is mandatory: taking the middle of the unsorted list (14) is the standard mistake.

INDEPENDENT SET

1. Find the mean of 7, 9, and 14.
2. The mean of six numbers is 12. What is their sum?
3. The mean of four numbers is 9. Three of them are 6, 8, and 10. Find the fourth.
4. The number 25 was added to five numbers, and the mean of the six became 15. What was the mean of the original five?
5. Find the median of 5, 12, 9, 3, 11, 7.
6. The average age of four children is 9. What will their average age be in 3 years?
7. Mom is 36, Dad is 38, and the kids are 4 and 6. What is the average age of the family?
8. The mean of twenty numbers is 10. One number equal to 29 is replaced by 9. What is the new mean?
9. ★ The average of five different positive integers is 10. What is the greatest possible value of the largest of these numbers?

Lesson 1.5. Estimation, Rounding, and the Last Digit

THEORY

Estimate instead of computing. On the AMC 8 the answer choices are often far apart: $49 \cdot 21$ is about $50 \cdot 20 = 1000$, and that is enough to pick the answer. A ten-second estimate beats a one-minute exact computation.

Round one factor up and the other down — the errors nearly cancel: $49 \cdot 21 \approx 50 \cdot 20 = 1000$ (exactly 1029).

The last digit. The last digit of a product depends only on the last digits of the factors: for $7 \cdot 8 \cdot 9$, take $7 \cdot 8 = 56$ — digit 6, then $6 \cdot 9 = 54$ — digit 4. This checks answers and solves problems where exact computation is hopeless. The last digits of powers repeat in a cycle: for powers of 3 they go 3, 9, 7, 1, 3, 9, ... — a cycle of length four.

Parity and divisibility as a check. The sum of two odd numbers is even; a product with an even factor is even; the sum of three consecutive numbers is divisible by 3. One parity check catches half of all arithmetic slips.

WORKED EXAMPLES

EXAMPLE 1 • ESTIMATION

Which is closest to $49 \cdot 21$: 800, 1000, 1200, 1500, or 2000?

→ $49 \cdot 21 \approx 50 \cdot 20 = 1000$; exactly 1029. Answer: **1000**. One round up, one round down — and the estimate is nearly exact.

EXAMPLE 2 • LAST DIGIT

What is the last digit of $7 \cdot 8 \cdot 9$?

→ $7 \cdot 8 = 56$ — digit 6; $6 \cdot 9 = 54$ — digit 4. (Exactly: 504.) No need to compute the product itself.

EXAMPLE 3 • DIVISION WITH AN ESTIMATE

How many *full* weeks are there in 2026 days?

→ $2026/7$: estimate $7 \cdot 280 = 1960$, leaving $66 - 9$ more weeks and 3 days. In all, **289 full weeks** ($289 \cdot 7 = 2023$, remainder 3).

EXAMPLE 4 • PARITY

Without adding them all, decide whether $1 + 2 + \dots + 10$ is even or odd.

→ There are five odd terms (1, 3, 5, 7, 9) — an odd count of odd numbers makes the sum odd: **odd** (indeed, 55).

INDEPENDENT SET

1. Round 4,872 to the nearest hundred.
2. Which is closest to $68 \cdot 102$: 600, 700, 6,000, 7,000, or 70,000?
3. What is the last digit of 3^4 ?
4. Compute $597 + 404$.
5. What is the last digit of $25 \cdot 25 \cdot 25$?
6. Compute $7 \cdot 11 \cdot 13$.
7. The sum of three consecutive positive integers is always divisible by what number?
8. Which is larger: 2^{10} or 10^3 ?
9. ★ What is the last digit of 7^{2026} ?

Solutions to the independent sets

LESSON 1.1

1. Divide by 6: $\frac{3}{4}$.
2. $5 \cdot 12 = 60$ is more than $7 \cdot 8 = 56$: $\frac{5}{8}$.
3. $\frac{5}{6} - \frac{3}{6} = \frac{2}{6} = \frac{1}{3}$.
4. $63/9 = 7$, then $7 \cdot 2 = 14$.
5. If $2/7$ is 10, then $1/7$ is 5, so the number is **35**.
6. $\frac{3}{6} + \frac{2}{6} + \frac{1}{6} = 1$.
7. $24 \cdot 3/8 = 9$.
8. $\frac{4}{12} < ? < \frac{6}{12}$: the answer is $\frac{5}{12}$.

★ Let there be N students: girls are $3N/5$, which also equals $(N + 4)/2$. Then $6N = 5N + 20$, so $N = 20$. Check: 12 girls of 20; now 24 students, and 12 is exactly half.

LESSON 1.2

1. $45/5 = 9$.
 2. $12/48 = 1/4 = 25\%$.
 3. $8 + 4 = 12$, so $80 + 12 = 92$.
 4. $7 \cdot 20 = 140$.
 5. 50 out of 250 is $50/250 = 1/5 = 20\%$. The percent is always taken of the OLD price.
 6. $200 \cdot 0.65 = 130$.
 7. $3 \cdot 12 = 36$.
 8. If $80\% = 96$, then $10\% = 12$, so the price was **\$120**. The trap is adding 20% to 96: that gives 115.20, not the original price.
- ★ We need to multiply $5/4$ by something to get 1: that is $4/5$, a drop of one fifth. $p = 20$. Percent up and percent down are not symmetric: +25 is undone by -20.

LESSON 1.3

1. One part is $45/9 = 5$: the parts are 10 and **35**.
 2. Divide by 6: **2 : 3**.
 3. One apple is 80 g, eight are **640 g**.
 4. $a : b = 6 : 8$ and $b : c = 8 : 20$, so $a : c = 6 : 20 = 3 : 10$.
 5. $180/3 = 60$ words per minute.
 6. Five times: $x = 3 \cdot 5 = 15$.
 7. One part is 7 oz, five parts in all: **35 oz**.
 8. One part = 3: nine parts in all, so **27**.
- ★ $(5k - 10) : 3k = 5 : 4$, so $20k - 40 = 15k$ and $k = 8$. The numbers are 40 and 24, sum **64**. Check: $30 : 24 = 5 : 4$.

LESSON 1.4

1. $30/3 = 10$.

2. $6 \cdot 12 = 72$.

3. The sum must be 36; we have 24: the fourth is **12**.

4. The old sum is $90 - 25 = 65$, mean $65/5 = 13$.

5. Sorted: 3, 5, 7, 9, 11, 12. The median is the average of the two middle values: $(7 + 9)/2 = 8$.

6. The mean grows by 3 as well: **12**.

7. $(36 + 38 + 4 + 6)/4 = 84/4 = 21$.

8. The sum dropped by 20: from 200 to 180. Mean: $180/20 = 9$.

★ The sum is 50. To maximize one number, make the others the smallest possible distinct values: $1 + 2 + 3 + 4 = 10$. The largest: $50 - 10 = 40$.

LESSON 1.5

1. **4,900**.

2. $\approx 7,000$ (exactly 6,936): **7,000**.

3. $3^4 = 81$: digit **1**.

4. $600 + 400 = 1000$, correction $-3 + 4 = +1$: **1001**.

5. A product of numbers ending in 5 ends in **5**.

6. $77 \cdot 13 = 1001$. Worth memorizing: $1001 = 7 \cdot 11 \cdot 13$.

7. The sum is $3n$: divisible by **3**.

8. $1024 > 1000$: **2^{10}** .

★ The cycle: 7, 9, 3, 1 (length 4). $2026 = 4 \cdot 506 + 2$ — the second position of the cycle: **9**.